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Does Structured Triage Context Help a Chest-Radiograph Pneumonia Model? A Time-Safe, Multi-Seed, Externally-Validated Multimodal Study with Missing-Data Cautions

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Abstract Background: Emergency-department (ED) triage records structured vital signs before chest imaging, motivating multimodal models that fuse the chest radiograph (CXR) with clinical context for pneumonia detection. Whether such fusion actually helps—under strict time-safety and rigorous evaluation—is unclear. **Methods:** We build a temporally safe cohort from MIMIC-CXR-JPG, MIMIC-IV and MIMIC-IV-ED, anchoring every feature at the CXR acquisition time t_0 and admitting only measurements resolved at or before t_0. On a patient-level temporal split (test n=1,075), we train image-only, concatenation- and attention-fusion, and clinical-baseline models across five random seeds, adding an early-laboratory panel as a third modality. We report AUROC, AUPRC, expected calibration error (ECE) and Brier score as mean±SD over seeds with paired per-seed deltas, external validation on NIH ChestX-ray14, and explicit leakage and missing-data audits. **Results:** A fine-tuned DenseNet-121 reaches AUROC 0.737 ± 0.003, far above triage-only baselines (0.61). Adding triage vitals leaves discrimination unchanged (ΔAUROC $+0.004\pm0.009$, within a pre-specified ±0.05 margin). A calibration advantage reported from a single checkpoint (ECE 0.040 vs 0.067) shrinks to ΔECE -0.013 ± 0.016 across seeds—single-seed evaluation overstated it $\sim2\times$. Adding early labs yields a small, seed-consistent AUROC gain ($+0.009 \pm 0.005$) that an ablation shows is reproduced exactly by lab *missingness indicators alone*, with no contribution from lab values, and is not attributable to leakage. Externally, discrimination transfers with minimal loss (AUROC 0.722 ± 0.005 on 112,120 NIH radiographs). **Conclusion:** The chest radiograph carries the discriminative signal; immediately available clinical context does not reliably add discrimination, and naive multimodal fusion can appear beneficial for reasons that will not generalise. We offer two transferable cautions for clinical multimodal AI: single-seed calibration inflation, and missingness masquerading as multimodal benefit.

Keywords Pneumonia detection · Chest radiograph · Multimodal fusion · Emergency department triage · Calibration · Reproducibility · Missing data · External validation

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1 Introduction

Pneumonia is a leading cause of ED presentation and admission, and the frontal chest radiograph (CXR) remains central to its diagnosis [Metlay et al(2019)Metlay, Waterer, Lon
Deep-learning CXR classifiers approach radiologist-level performance on several findings, and a substantial literature proposes fusing imaging with structured clinical data [Acosta et al(2022)Acosta, Falcone, Rajpurkar, and Topol, Hayat et al(2022)Hayat, Geras, and Shamout,Soenksen et al(2022)S

Three methodological issues recur in that literature and motivate this study. First, *temporal safety*: clinical variables are frequently included without verifying they were available at prediction time, risking optimistic leakage. Second, *single-seed reporting*: calibration and small discrimination effects are often reported from a single training run, although both are sensitive to training stochasticity [Guo et al(2017)Guo, Pleiss, Sun, and Weinberger, Nixon et al(2019)Nixon, Dusenberry, Zhang, Jerfel, and Tran]. Third, *missing data*: clinical inputs in the ED are heavily and non-randomly missing, so a model can exploit *whether* a measurement exists rather than its value.

We address a deliberately conditional question—*when, if ever, does immediately available clinical context help a CXR pneumonia model?*—under strict time-safety, with multi-seed replication and explicit missing-data and leakage audits, reported in line with prediction-model and medical-imaging guidance [Collins et al(2024)Collins, Moons, Dhiman et al, Tejani et al(2024)Tejani, Klontzas, Gatti, Mongan, Moy, Park, and Kahn]. Our contribution is not a new architecture but a rigorous, reproducible, externally-validated evaluation and two transferable cautions for clinical multimodal fusion: (i) single-seed evaluation can overstate a calibration advantage roughly twofold; and (ii) a multimodal model can present a spurious "benefit" driven by missing-data structure rather than the measured values.

2 Materials and Methods

2.1 Data and cohort

We link MIMIC-CXR-JPG v2.1.0 [Johnson et al(2019)Johnson, Pollard, Berkowitz, Greenbaum, Lungren, Deng, Mark, and Horng], MIMIC-IV v2.2 [Johnson et al(2023b)Johnson, Bulgarelli, Shen, ...] and MIMIC-IV-ED v2.2 [Johnson et al(2023a)Johnson, Bulgarelli, Pollard, Horng, Celi, and Mark]. The anchor t_0 is the CXR acquisition timestamp (DICOM StudyDate+StudyTime). Studies are restricted to frontal ED-anchored radiographs with a linkable ED stay. Pneumonia labels are the CheXpert label [Irvin et al(2019)Irvin, Rajpurkar, Ko et al] from the radiology report; radiologically uncertain studies are excluded (`u_ignore`). The cohort comprises 9,154 ED-anchored studies; the patient-level temporal split assigns train 7,144 / validation 930 / test 1,080 studies, and models train and evaluate on radiographs available on disk, yielding a held-out test set of n=1,075 (487 positive / 588 negative; prevalence 45.3%).

2.2 Temporal safety and split

Every clinical feature is a triage measurement or a laboratory value resolved at or before t_0; for labs we additionally require charttime within $(t_0-24\,\mathrm{h}, t_0]$. Time-safety was verified empirically: of 314,145 extracted lab measurements, zero have charttime after t_0. Patients are ordered by their first t_0 and assigned 80/10/10 at the patient level; no patient appears in more than one split (0 cross-split subjects, 0 duplicate studies).

2.3 Models

The image model is DenseNet-121 [Huang et al(2017)Huang, Liu, van ... ImageNet-initialised, multilabel-pretrained on 182,637 non-ED MIMIC-CXR studies, then fine-tuned for binary pneumonia (224 px, class-balanced BCE). Multimodal variants concatenate image features with a tabular MLP over triage features (vitals, acuity, view, demographics, per-vital missingness flags), or fuse them via a small Transformer (attention ablation). The +labs variant appends 26 early-lab values and their missingness flags. Clinical baselines are logistic regression and XGBoost [Chen and Guestrin(2016)] on triage features. A modern-backbone baseline uses a CheXpert-trained DenseNet-121 [Cohen et al(2022)Cohen, Viviano, Bertin, Morrison, ... zero-shot and as a frozen-feature linear probe. All deep models share the pretrained backbone so the multi-seed analysis isolates fine-tuning stochasticity; tabular preprocessing (median imputation, standardisation, one-hot) is fit on training data only.

2.4 Evaluation

Primary metrics are AUROC, AUPRC, ECE (10 uniform bins) [Guo et al(2017)Guo, Pleiss, Sun, and Weinberger] and Brier score. Each deep model is trained across five seeds $\{42, 123, 456, 789, 1000\}$; we report mean±SD and paired, per-seed deltas against the image-only model on identical test rows. Within-checkpoint uncertainty uses a 2,000-replicate patient-level paired bootstrap. A non-inferiority margin Δ_0=−0.05 AUROC was prespecified [Piaggio et al(2012)Piaggio, Elbourne, Pocock, Evans, and A... Additional analyses: ECE bin-count/scheme sensitivity, operating points at fixed sensitivity, subgroup metrics, a missingness-only labs ablation, and external validation on all NIH ChestX-ray14 radiographs [Wang et al(2017)Wang, Pe... with preprocessing matched to the internal evaluation transform.

3 Results

3.1 The image dominates; fusion is discrimination-neutral

Across five seeds (Table 1, Fig. 1), both CXR models beat triage-only baselines by more than 12 AUROC points. Adding triage vitals does not change discrimination (concat − image ΔAUROC $+0.004 \pm 0.009$; lower bound well inside the -0.05 non-inferiority margin). Attention fusion matches image AUROC but is severely and variably miscalibrated (ECE 0.076 ± 0.042; Fig. 2).

3.2 Caution 1: single-seed evaluation inflates the calibration effect

A single checkpoint suggested the concat model was markedly better calibrated (ECE 0.040 vs 0.067, a 40% reduction). Across seeds the image-only ECE alone ranges 0.041–0.060, and the paired advantage shrinks to ΔECE -0.013 ± 0.016 (4/5 seeds favour concat; range crosses zero). The single-seed estimate overstated the benefit roughly twofold; the effect is also binning-dependent (Fig. 6) [Nixon et al(2019)Nixon, Dusenberry, Zhang, Jerfel, and Tran]. Brier scores are essentially unchanged, so any effect is distributional alignment, not refinement.

3.3 Caution 2: the labs "benefit" is missing-data structure

Appending early labs gives the only seed-consistent AUROC gain (labs − image $+0.009 \pm 0.005$, 5/5 seeds, range excludes zero). Three analyses show this is neither chemistry nor leakage. *Ablation:* a model using lab *missingness flags only* (no values) reproduces the gain exactly (flags − image +0.009; Fig. 2); lab values add nothing further to discrimination. *Subcohort:* on the 269 test studies that actually have labs, discrimination is tied (labs 0.778 vs image 0.776). *Leakage audit:* 0 of 314,145 lab measurements post-t_0; 0 cross-split subjects; preprocessing fit on train only; the "any lab drawn" indicator has near-null univariate AUROC (0.464). The gain thus reflects the weak complementary signal of *whether* bloodwork was ordered, is small, and lies within the patient-level bootstrap CI.

3.4 Clinical availability of labs

Under time-safe matching, only 25.4% of ED-anchored CXR studies (272/1,075 test) have any resolved lab at or before imaging; CBC and metabolic panels reach ∼25%, but CRP and procalcitonin—the most pneumonia-specific markers—are present in 0.5% and 0.6% (Fig. 4). At the moment of imaging, the informative labs are essentially unavailable.

3.5 Operating points, subgroups, modern backbone

At a fixed sensitivity of 0.90, specificity is 0.31 (image) vs 0.32 (concat); at 0.95, 0.17 vs 0.18—clinically equivalent. Subgroup AUROC reveals disparities not closed by fusion (Fig. 5): male 0.77 vs female 0.71; White 0.75, Black 0.71, Hispanic 0.64 (small strata); PA view 0.75 vs AP 0.72 [Seyyed-Kalantari et al(2021)Seyyed-Kalantari, Zhang, M Larrazabal et al(2020)Larrazabal, Nieto, Peterson, Milone, and Ferra A CheXpert-trained DenseNet-121 [Cohen et al(2022)Cohen, Viviano used zero-shot (AUROC 0.667) or as a frozen-feature linear probe (0.687), does not outperform our task-tuned model (0.737).

3.6 External validation

We evaluated the image models on all 112,120 radiographs of NIH ChestX-ray14 [Wang et al(2017)Wang, Peng, Lu, Lu, B (a different institution, scanner population and label pipeline; 1,431 pneumonia-positive, prevalence 1.28%). Discrimination transfers with minimal loss: AUROC 0.722 ± 0.005 across five seeds (ensemble 0.726), only 0.016 below the internal 0.737 (Fig. 3). As expected under the large prevalence shift (45%→1.3%), prevalence-dependent metrics degrade (AUPRC 0.030, ECE 0.44) because probabilities are calibrated to the internal base rate; AUROC (prevalence-independent) is the appropriate cross-dataset metric, and it holds despite domain shift and NLP-derived external labels.

4 Discussion

The chest radiograph carries the discriminative signal for radiographic pneumonia in this ED cohort; immediately available structured clinical context—triage vitals, and even a broad early-lab panel—does not add reliable discrimination, and the image model generalises externally with minimal loss. This is a well-powered negative answer to a specific, clinically motivated question, accompanied by two transferable methodological cautions. Reporting calibration or small discrimination effects from a single training run can overstate them (here ∼2×); multi-seed replication should be standard. And where clinical inputs are heavily and non-randomly missing, a multimodal model can exploit missingness structure and present a spurious benefit that neither reflects the measured values nor generalises—a failure mode distinct from data leakage, which our audits explicitly excluded.

Limitations. Training and multimodal evaluation are single-institution (MIMIC) and retrospective; external

Table 1 Multi-seed test-set performance (mean±SD over 5 seeds; n=1,075; prevalence 45.3%). Clinical baselines are single-fit.

Model	AUROC	AUPRC	ECE	Brier
Image-only DenseNet-121	0.737 ± 0.003	0.719 ± 0.004	0.053 ± 0.008	0.207 ± 0.001
Multimodal — concat (triage)	0.741 ± 0.006	0.715 ± 0.005	0.040 ± 0.014	0.205 ± 0.002
Multimodal — attention (triage)	0.738 ± 0.006	0.713 ± 0.012	0.076 ± 0.042	0.212 ± 0.010
Multimodal — concat (+labs)	0.747 ± 0.004	0.724 ± 0.004	0.043 ± 0.011	0.203 ± 0.002
Logistic regression (triage)	0.606	0.548	0.037	0.242
XGBoost (triage)	0.611	0.567	0.046	0.241

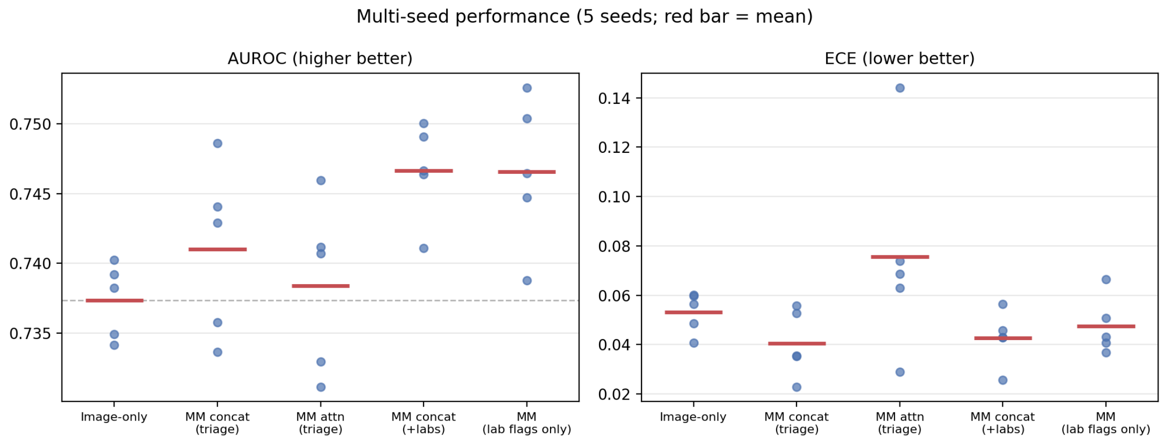

Fig. 1 Per-model AUROC and ECE across five seeds (points, seeds; red bar, mean). Discrimination is stable; the image-only ECE alone varies substantially by seed.

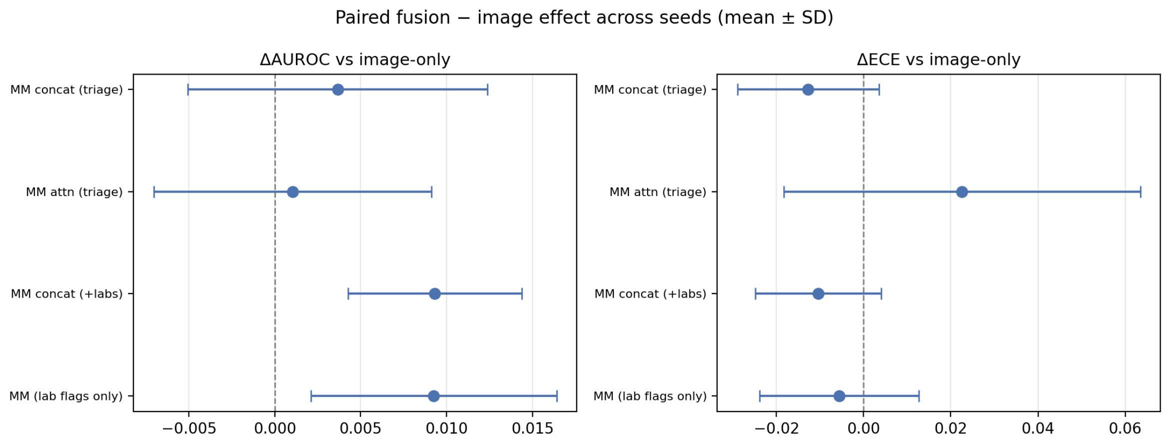

Fig. 2 Paired fusion − image effects across seeds (mean±SD). Every effect straddles zero; the labs and lab-flags-only deltas are identical, indicating the labs effect is driven by missingness rather than lab values.

validation confirms the image branch generalises to NIH but is necessarily limited to imaging, as no external dataset links ED triage to the CXR, and probabilities require recalibration to the target base rate before deployment (the `u_ignore` policy also raises internal evaluation prevalence to 45.3%). The lab-present subcohort is small (n=269); the modern-backbone comparison uses frozen features rather than end-to-end fine-tuning; and NIH labels are NLP-derived, adding label noise to the external estimate.

5 Conclusion

Under strict time-safety and multi-seed evaluation, fusing immediately available clinical context with the chest radiograph does not reliably improve pneumonia discrimination, and the one apparent gain is an artefact of missing-data structure. The image model itself is robust and generalises across institutions. Beyond the negative result, the study offers two reusable cautions—single-

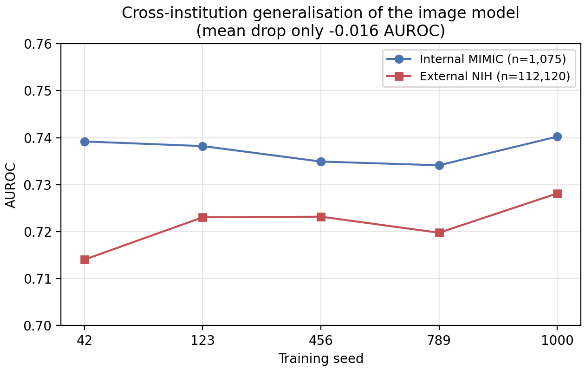

Fig. 3 Cross-institution generalisation: per-seed AUROC on internal MIMIC (n=1,075) vs external NIH ChestX-ray14 (n=112,120); mean drop −0.016.

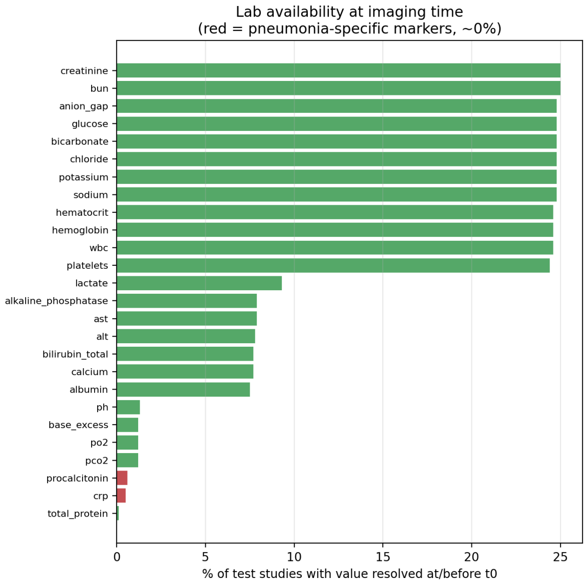

Fig. 4 Fraction of test studies with each lab resolved at or before t_0; pneumonia-specific markers (red) are essentially unavailable at imaging time.

seed calibration inflation and missingness-as-signal—for the growing field of clinical multimodal AI.

Declarations

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Authors' contributions. YAD designed and implemented the data pipeline, trained all models, and performed the analyses. AME conceptualised and supervised the

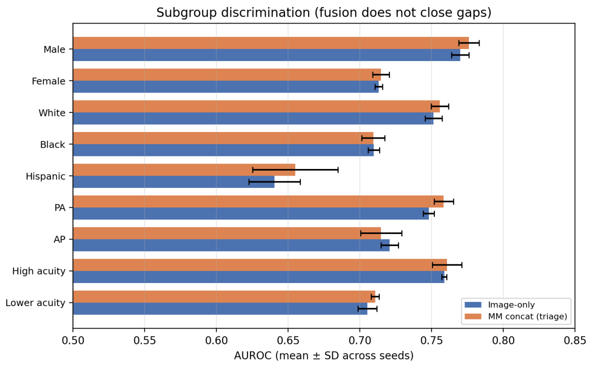

Fig. 5 Subgroup AUROC (mean±SD across seeds) for image-only vs concat fusion; fusion does not close sex, race or view gaps.

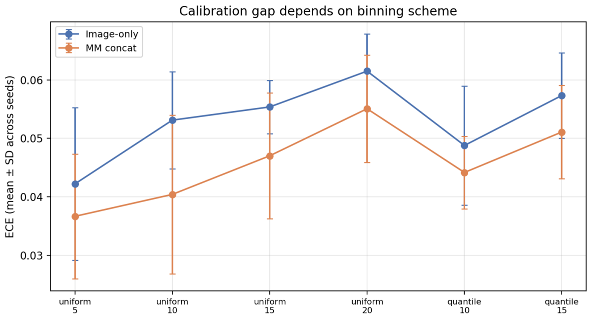

Fig. 6 ECE under different binning schemes; the image–concat calibration gap depends on the binning choice.

study and contributed to the manuscript. MM and SS contributed to methodology, interpretation, and critical revision. All authors read and approved the final manuscript.

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Conflict of interest. The authors declare no competing interests.

Ethics approval. All data are publicly available and de-identified (MIMIC under PhysioNet's credentialed Data Use Agreement; NIH ChestX-ray14 public release). No additional ethical approval was required; procedures followed the Declaration of Helsinki.

Data and code availability. MIMIC and NIH ChestX-ray14 are publicly available (MIMIC via PhysioNet credentialing). All pipeline, training, and evaluation code is available at `https://github.com/wa5rx3/MultiModalPneumonia`. Every reported number is regenerated by a named script tied to the fixed temporal split.

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